

Pallavi Desai

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EDUCATION

University of Texas at Austin

M.S. in Computer Science • College of Natural Sciences • **Relevant Coursework:** Generative Models, Databases and LLMs, Machine Learning on Real World Networks, Grounded NLP, Distributed Computing. **Thesis Advisor:** [Dr. Risto Miikkulainen](#).

University of California, Berkeley

B.A. in Data Science Class of 2025 • College of Computing, Data Science, and Society • SEED Scholars Honors Program • Minor in Science, Technology, & Society • Certificate in Entrepreneurship and Technology from the Sutardja Center • **Relevant Coursework:** Data Engineering; Inferencing, Modeling, & Decision Making; Probability for DS; Data Structures & Algorithms.

PATENTS/PUBLICATIONS

Towards Interpretable Models of Rumor Spread in LLM-Driven Agent Societies

August-December 2025

- Collaborated with [Rahul Nandakumar](#), PhD student in the McCombs School of Business, to examine how rumors spread and what network features accelerate or hinder them using graph theory and LLM-agent simulations. Will be submitted for publication in summer 2026.

Independent Project: 2 U.S. Patents

January 2020-Present

- 2 issued patents ([#10,885,891](#)) for AIMarchAssist technology: a thin, lightweight, electronic display for sheet music that uses Real-Time Kinematic tracking and AI-based telemetric assessment to provide marching bands with **individualized, real-time feedback** for music and marching performance. Continuation patent ([#11,289,058](#)).

EXPERIENCE

Machine Learning Engineer Intern, American Express

June-August 2024

- Conditioned 1.5 years worth of Slack messages for an intent classification model for AmEx's Google Cloud Platform. Trained a Multinomial Naive Bayes Model and fine-tuned hyperparameters to **improve model accuracy from 22% to 61%**.
- Verified results using K-Fold Cross Validation. Built training dataset of **120 keyword and intent pairs** from scratch.
- **Effectively communicated highly technical findings** to a non-technical SVP-level audience through business implications.
- Later presented a poster on this work at the 2024 SEED Scholars Honors Program Research Symposium.

Founder and CEO, AIMarchAssist

August 2023-Present

- Led team of 2 to develop above [patented](#) technology behind marching band startup, **1** of 22 selected out of 150 for Berkeley's [Student Entrepreneurship Program](#). Calculated TAM and SAM to date. Researched gaps in the antiquated market, product-market fit was overwhelmingly positive, but low budget cost effectiveness is important.
- Managed interview pipeline to recruit **50+** customers/users by proactively cold-emailing student groups and band directors.

Data Scientist Intern, American Express

June-August 2023

- Synthesized 4 years' worth of user feedback from an internal Slack channel to consolidate feedback to improve features on the Cornerstone Interactive Analytics platform to better understand pain points.
- Designed/implemented a lexicon-based sentiment analysis algorithm using NLTK's Vader to create a Bag of Words Model for 20,000 lines of feedback that gives **accurate sentiment scores and word frequencies**.
- Enhanced consumer insights by providing an overview of current issues for 14 Product Managers in the business unit.

Product Management Intern, [You.com](#)

June-August 2022

- Managed and modernized the Social Search feature (search results from social media) of [privacy-based LLM aggregator](#).
- **Increased user retention & conversion** through customer behavior analysis and performance prediction.
- Strengthened 3 Product Requirements Documents, coordinated tasks in Product, Engineering, Design, Marketing, and Analytics.
- Made 15-20 feature enhancements to Social Search based on 50+ user interviews, leading to increased traffic on Social Search by **159% in 2 months**. Presented to **C-suite and implemented feedback** so my contributions were backed by customer feedback.

RESEARCH

Graduate Research Assistant, [DynamCog Lab](#)

January 2026-Present

- Collaborated with [Dr. Nicholas Duran](#) and another student to analyze secure messages between patients and healthcare providers to improve patient-centered outcomes and enhance comparative effectiveness research dissemination in practice.
- Utilized LLMs to conduct Named Entity Recognition on a synthetic dataset to remove personally identifiable information. In collaboration with the University of California, San Francisco and Kaiser Permanente.

Fung Fellow, Fung Institute for Engineering Leadership

August-December 2024

- Partnered with [Reclaim Our Power](#), a grassroots campaign advocating for utility justice, to develop an interactive GIS-based virtual resource that addresses high utility costs and environmental dangers by PG&E to taxpayers.
- Applied design principles and expanded creativity skills to create **dynamic visualizations** using arc-GIS software in group of 4.

Research Assistant, Trademark Rebranding Project

September-December 2022

- Collaborated with a team led by [Prof. Sonia Katyal](#) to synthesize **12-column data** on why public companies choose to rebrand from a series of 3,000 articles over the last 20 years.
- Assembled a subset of over 150 articles from the last 12 years to **conduct NLP analysis** to determine rebranding motives.

Research Assistant, Micro Lab at Haas Business School

January-May 2022

- Examined online vs. in-person interaction, the nonverbal detection of racial bias, the tendency to delay gratification, and other projects, including coding nonverbal behaviors for [Dr. Dana Carney](#). Investigated why powerful people are better liars and the outcomes of power and stress through a **preliminary literature review** of 10 articles from 2012 or later related to these topics.

LEADERSHIP

Incoming Co-President, UT [Graduate Representative Association of Computer Sciences](#) (GRACS)

August 2025-Present

- Collaborated with 15 students to host events such as Tea Time, happy hours, faculty Q&A sessions, and other events. Started on the First Year Liaison Team of 2 that works to ensure a smooth transition to the UTCS community for incoming students.
- Created [UTCoffeeChat](#), a secure, online platform with 20+ users to encourage mentorship between senior and first year MS/PhD students and cross-pollination with ECE and Geosciences.
- Currently working with 3 other students to create an MSCS 101 Advice Page for incoming first year students.
- Conducted a Climate Survey to assess student feelings about department policies and presented results to [Dr. Peter Stone](#).
- Elected as Co-President starting in Fall 2026, previously served as the First Year Liaison Officer.

Senior Mentor, Berkeley Mentorship Cohort

August 2022-May 2025

- Mentored 2 freshmen per year for 3 years providing **individualized help** on adjusting to Berkeley, classes, LinkedIn, applying to research and internships, exam studying, organization skills, etc.
- Adapted mentorship style to guide my mentees, held weekly one-on-one meetings and check-ins, invited them to upcoming events, **exhibited empathy** as they worked through difficult situations. [Program website](#).

Operations Manager and Teaching Assistant, Sutardja Center for Entrepreneurship and Technology

August 2024-May 2025

- Collaborated with 8-9 people to **coordinate workshops** for the SCET Innovators in Residence program and community as a **Peer Startup Coach** on themes such as problem discovery, startup therapy, pitching, and storytelling.
- **Taught** 13 students at the SCET Startup Ideation and Ecosystems Workshop to solve the [UN's Sustainable Development Goals](#). **Designed workshop curriculum** and planned activities from scratch. For additional context, [SCET IIR website](#).
- Workshops were consistently ranked as the most enjoyed activity by the SCET Innovation Fellows program.

Senior Member, College of Computing, Data Science, and Society Student Advisory Board

August 2023-May 2025

- Appointed to the new College of CDSS's Undergraduate Student Advisory Board to help **craft policies** and regulations.
- Planned orientation activities and pitched college-wide **mentorship program** for incoming freshmen/transfers. **Provided feedback** to administrators on policies concerning enrollment, requirements, breadth classes, advising, and office hours.

Competitive Member, Cal Equestrian Team

August 2022-July 2025

- **Mentored new riders** on **horsemanship skills** and techniques, **mental exercises**, and muscle building.
- Placed **3rd out of 11** at UC Santa Cruz [Intercollegiate Horse Show Association](#) (IHSA) competition (November 2024).
- Cal Equestrian consistently received **Reserve Champion (2nd place)** at Fall 2022 and 2024 competitions out of 6 teams.
- Completed a month-long [training program](#) in Cordoba, Argentina in July 2025 and placed **3rd out of 15** in a local competition.

HONORS AND AWARDS

SEED Scholar, STEM Excellence Through Equity and Diversity Honors Program

July 2021-May 2025

- Engaged in a 6-week [Summer Rising](#) program in July 2021, which entailed taking courses in **Data Science**, Algebra, Psychology, and Biology. Fostered in-depth collaboration and discussion with 25 scholars on the importance of DE&I in STEM.
- Built collaborative relationships with sister programs, graduate mentors, other SEED cohorts, and **faculty members**.

Most Promising Innovation Award, Designing Startups to Transform Society Pitch Competition

May 2024

- Collaborated with a team of 5 to conceptualize an idea for a personal data tracking app via speech or text input. Refined and substantiated the concept through **30+ comprehensive customer validation interviews**. Supervised by [Dr. Anita Balaraman](#).
- Created and pitched a business model which received the **Most Promising Innovation Award** out of 13 teams.

Community Award, Berkeley Mentorship Cohort

April 2023

- Awarded to **1 out of 95** mentors in the 2022-2023 academic year for an extraordinary demonstration of **equity and inclusion values** within the College of Engineering community.
- Presented by [Dean Tsu-Jae King Liu](#) and Program Director Marcia Breslin-Cantillana. Nominated by one of my mentees.

SKILLS

- **Technical Skills:** Python, SQL, Java, MongoDB, Git, Numpy, SciPy, Matplotlib, SciKit Learn, NLTK, SpaCy, etc.
- **Languages:** Nepali, Spanish, Hindi, Gujarati.
- **Interests:** flute musician, reading, aviation, photography, live music, cooking, and traveling.